MOBILE APPLICATION DEVELOPMENT(CS-575)

PROJECT DOCUMENTATION

INTERACTIVE HABIT GARDEN APP

PROJECT TEAM:

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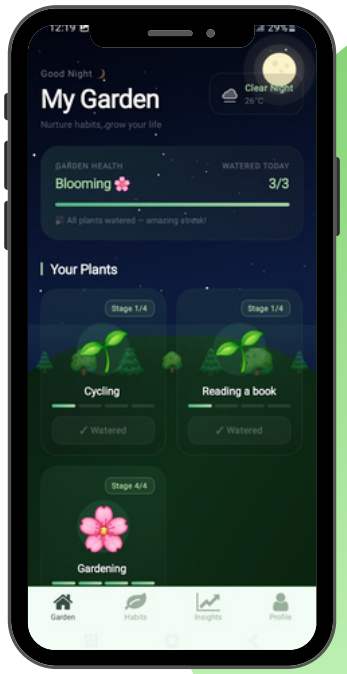

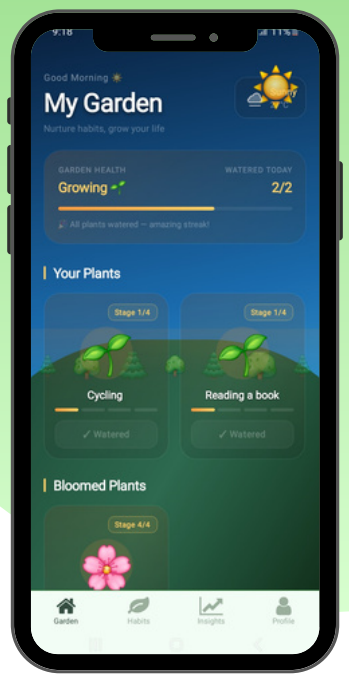

ACADEMIC OVERVIEW:

YEAR & SEMESTER: ***3rd YEAR & 5th SEMESTER***

SECTION: ***BS Computer Science – Section B***

COURSE-INSTRUCTOR: ***HABIB MUHAMMAD SABIR***

DEPARTMENT OF COMPUTER SCIENCE
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HABIT GARDEN APPLICATION DOCUMENTATION

Interactive habit tracking mobile application

App Overview

Habit Garden is a gamified habit-tracking mobile application. Users plant virtual seeds that grow into thriving plants as they maintain daily habits, blending the satisfaction of streaks with the visual joy of a digital garden. The metaphor of nurturing plants makes habit-building tangible, rewarding, and enjoyable.

Key Features

The application offers a range of features designed to engage users and encourage consistent habit tracking:

Visual Habit Garden (Garden Screen)

The centrepiece of the app is a living garden view where each habit is represented as a growing plant. The garden evolves visually as users log progress, providing an at-a-glance motivation dashboard.

- Each habit is rendered as a plant emoji that upgrades through growth stages
- Plants progress from Seedling → Sprout → Tree → Bloom
- Fully bloomed plants are highlighted with a Bloomed header
- Tap any plant card to navigate to the detailed Habit View

Habit Management (Habits Screen)

Users can create, track, water, and delete habits through an intuitive list interface with swipe-friendly cards.

- Add new habit with a name and goal duration in days
- Water (log) a habit once per day with a tap — enforces single daily check-in
- Visual progress bar and percentage for each habit
- Delete a habit with a confirmation dialog
- Navigate to a detailed breakdown screen per habit

Add Habit Screen

A dedicated screen for planting a new habit seed, featuring real-time visual feedback.

- Animated growth preview that reacts to the duration input
- Floating leaf particle animations for a playful feel
- Duplicate name validation before saving
- Saves the habit to both Redux state and Firebase Firestore

Habit Detail Screen

A rich analytics view for a single habit, showcasing journey progress.

- Hero gradient header with animated plant avatar
- Animated progress bar (animated from 0 to actual % on mount)
- Streak and goal day counters in stat cards
- Consistency trail: a visual dot-grid showing completed vs remaining days
- Dynamic status card: "Goal Achieved", "On Track", or "Keep Growing"

Insights Screen

A data-driven statistics screen that provides an analytical overview of all habits.

- Average progress ring (approximated CSS border trick)
- Total habits, fully-bloomed, growing, and seedling counts
- Best streak and longest goal pills
- Animated vertical bar chart — top 6 habits by streak
- Status breakdown (Bloomed / Growing / Seedling) with emoji
- Empty state when no habits have been added yet

Profile Screen

Displays user account information and app personalisation options.

- Displays the gardener's name, email, and profile avatar
- Theme switcher to personalise the app colour palette
- Includes quick tips accessible via the drawer navigation
- Allows enabling/disabling streak alert and reminder notifications via the drawer navigation
- Logout button that clears Redux state and returns to the Login screen

Motivational Quotes

- API-powered dynamic quotes refreshed daily
- Displayed contextually within the garden experience
- Encourages positivity and consistency

Animations & Micro-Interactions

- Smooth transitions and animated components throughout the UI
- Visual feedback when completing (watering) a habit
- Staggered card entrances and floating particle effects

- Enhances engagement and overall app feel

Real-Time Weather Background

- Dynamic garden background powered by live weather data using a weather API (e.g., OpenWeather API)
- Environment adapts to real-world conditions: sunny, rainy, cloudy, night, and evening
- Animated elements include twinkling stars, fireflies, falling rain, and a moon/sun
- Creates an immersive, ever-changing experience that mirrors the real world

Authentication Flow

Secure user authentication managed by Firebase Authentication, with a custom gate component.

- Login screen with email and password
- Sign-up screen accessible from within Login
- LoginGate component wraps the entire app — renders Login or the Tab Navigator depending on Redux auth state
- On login, user habits are fetched from Firestore and loaded into Redux
- Persistent session during the app lifecycle

Application Screens

Screen / File	Description
gardenscreen.tsx	Visual garden dashboard — plant cards overview
index.tsx (Habits)	Habit list with water, detail, and delete actions
add.tsx	Add new habit with animated preview
habitDetail.tsx	Full analytics view for a single habit
explore.tsx (Insights)	Aggregated statistics and charts
profile.tsx	User profile, theme, drawer nav, and logout
login.tsx	Email/password login form
signup.tsx	New account registration
LoginGate.tsx	Auth guard — gates entire tab navigator
_layout.tsx (root)	App root: Redux Provider + ThemeProvider + LoginGate
_layout.tsx (tabs)	Tab navigator with 4 bottom tabs

Technologies Used

Layer	Technology
Framework	React Native + Expo
Navigation	Expo Router (file-based routing)
State Management	Redux Toolkit (habits, weather, user slices)
Backend / Auth	Firebase Authentication + Firestore
Weather Data	WeatherAPI.com (live conditions + temperature)
UI & Animation	LinearGradient, BlurView, React Native Animated API
Icons	Expo Vector Icons (FontAwesome)
Language	TypeScript (95%) + JavaScript (5%)

Run Instructions

To run the Habit Garden application locally, follow these steps:

Step 1 — Clone the Repository

```
git clone https://github.com/neharehan2005/HabitBloom-Interactive-Habit-Garden-App.git

cd HabitBloom-Interactive-Habit-Garden-App
```

Step 2 — Install Dependencies

```
npm install
```

Step 3 — Configure Firebase

Open and replace the placeholder values with your Firebase project credentials:

```
const firebaseConfig = {
 apiKey: 'YOUR_API_KEY',
 authDomain: 'YOUR_PROJECT.firebaseapp.com',
```

```
  projectId: 'YOUR_PROJECT_ID',
  storageBucket: 'YOUR_PROJECT.appspot.com',
  messagingSenderId: 'YOUR_SENDER_ID',
  appId: 'YOUR_APP_ID'
};
```

Step 4 — Start the Development Server

```
npx expo start
```

Step 5 — Open on Device or Emulator

- Scan the QR code with the Expo Go app on iOS or Android
- Press to open in the iOS simulator (requires Xcode on macOS)
- Press to open in the Android emulator (requires Android Studio)
- Press to open in a web browser (limited feature support)

How It Works

HabitBloom follows a simple but powerful loop:

Step	Action	Visual Feedback
1	Plant a seed — create a habit with a name and goal duration	A seedling appears in your garden
2	Water daily — mark the habit complete each day	Progress bar fills up
3	Watch it grow — streak builds through 4 visual stages	Plant visually evolves
4	Bloom — reach 100% of your goal	Plant fully blooms

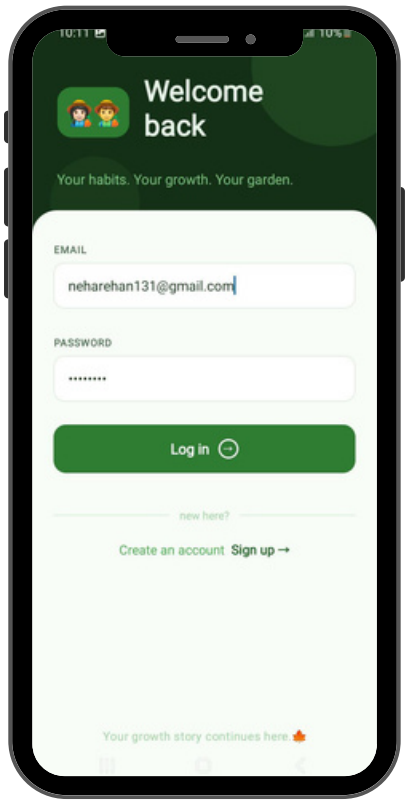
Repository

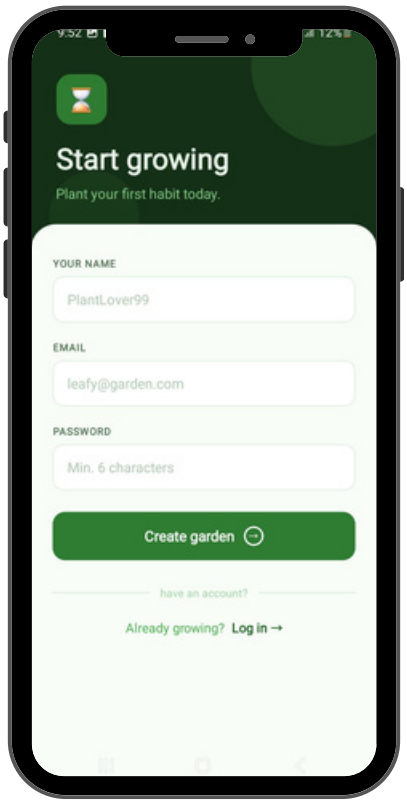
GitHub: **https://github.com/neharehan2005/HabitBloom-Interactive-Habit-Garden-App**

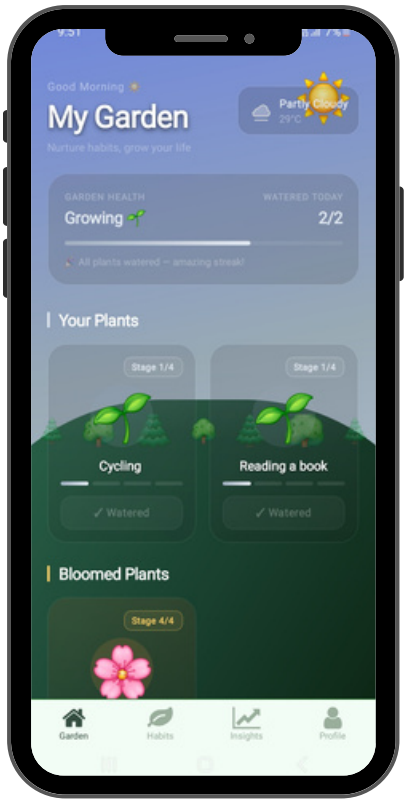
APK File

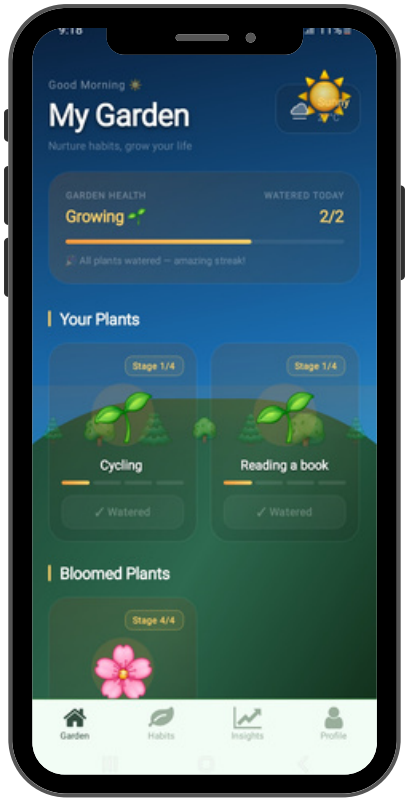
application-09f25bd4-9b66-4139-904b-b99d9835f77f-1.apk

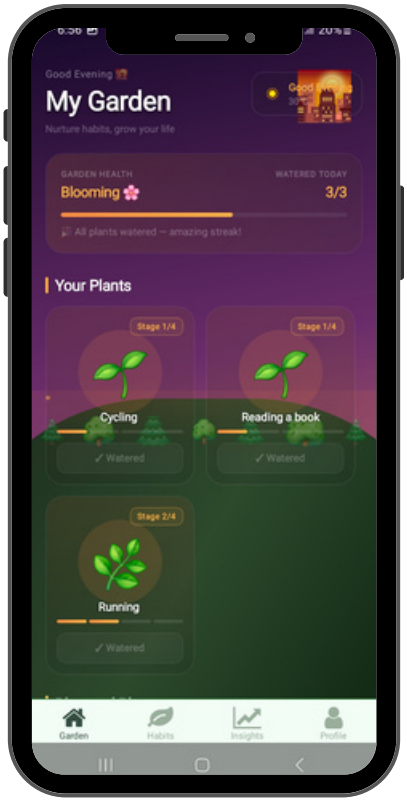
Screenshots

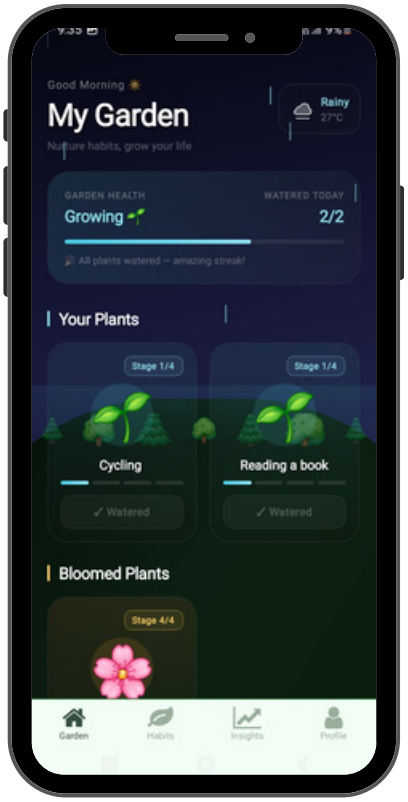


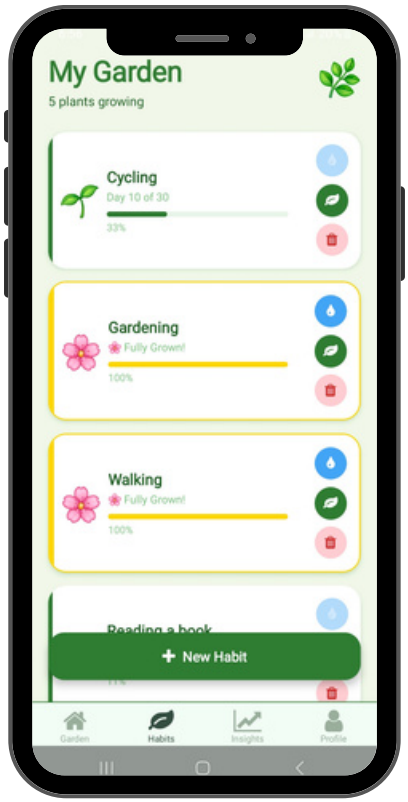


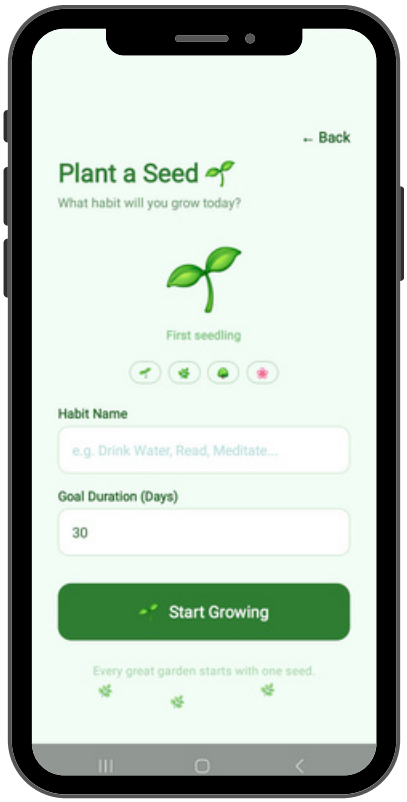



Good Morning
My Garden
Nurture habits, grow your life
GARDEN HEALTH
Growing
WATERED TODAY
2/2
All plants watered — amazing streak!
Your Plants
Stage 1/4
Cycling
✓ Watered
Stage 1/4
Reading a book
✓ Watered
Bloomed Plants
Stage 4/4
Garden
Habits
Insights
Profile

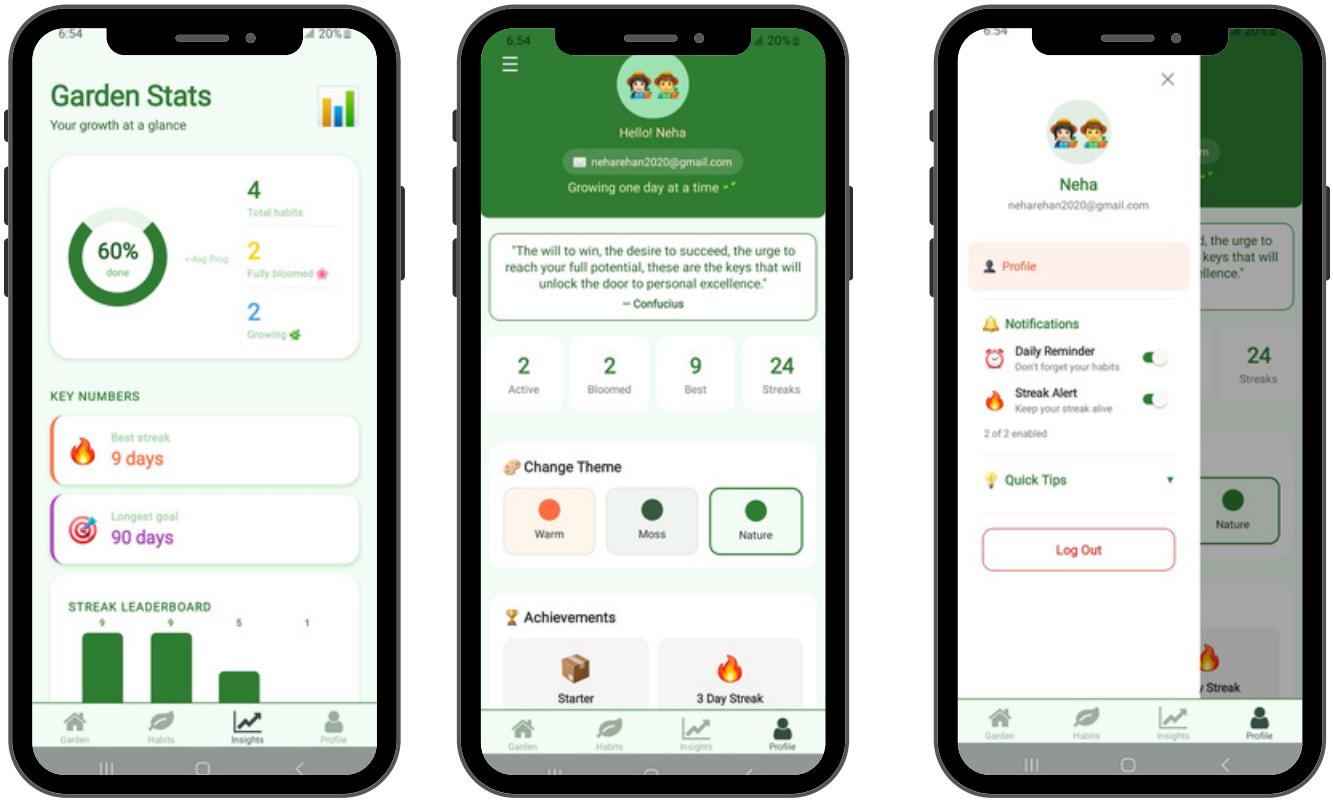

Good Evening
My Garden
Nurture habits, grow your life
GARDEN HEALTH
Blooming
WATERED TODAY
3/3
All plants watered — amazing streak!
Your Plants
Stage 1/4
Cycling
✓ Watered
Stage 1/4
Reading a book
✓ Watered
Stage 2/4
Running
✓ Watered
Garden
Habits
Insights
Profile


Good Morning
My Garden
Rainy
27°C
Nurture habits, grow your life
GARDEN HEALTH
Growing
WATERED TODAY
2/2
All plants watered — amazing streak!
Your Plants
Stage 1/4
Cycling
✓ Watered
Stage 1/4
Reading a book
✓ Watered
Bloomed Plants
Stage 4/4
Garden
Habits
Insights
Profile


My Garden
5 plants growing
Cycling
Day 10 of 30
33%
Gardening
Fully Grown!
100%
Walking
Fully Grown!
100%
Reading a book
+ New Habit
Garden
Habits
Insights
Profile


← Back
Plant a Seed
What habit will you grow today?
First seedling
Habit Name
e.g. Drink Water, Read, Meditate...
Goal Duration (Days)
30
Start Growing
Every great garden starts with one seed.

SPROUTING
Cycling
HABIT JOURNEY
Growth Progress
33%
10 of 30 days complete
20 days left
10
DAY STREAK
30
DAY GOAL
Consistency Trail
Completed
Remaining



Summary

This document provides a comprehensive overview of the Habit Garden mobile application, covering its features, technologies, setup instructions, and workflow.